

Partial Solutions To Exercises In Calculous 3

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Intro

Partial solution to exercises from 26A calc3 homeworks. Learned using Shiri's 25A course notes and Boris Tsirelson lectures from 2016.

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..... (1)

1. Suppose $A \in \leftrightarrow (\mathbb{R}^n, \mathbb{R}^m)$ and $B \in \leftrightarrow (\mathbb{R}^n, \mathbb{R}^m)$ are matrices. We'll prove HS norm and the operator norm a sub-multiplicative.

Proof. • **Operator norm:** Let $x \in \mathbb{R}^n$. Then by the very definition of $\|A\|_{op}$ and $\|B\|_{op}$ we get that:

$$\|ABx\| \leq \|A\| \cdot \|Bx\| \leq \|A\| \cdot \|B\| \cdot \|x\|$$

Hence we get that:

$$\forall x \in \mathbb{R}^n: \frac{\|ABx\|}{\|x\|} \leq \|A\|_{op} \cdot \|B\|_{op}$$

Therefore, the supremum on all $x \in \mathbb{R}^n$, which is defined to be $\|AB\|_{op}$, is bounded by $\|A\|_{op} \cdot \|B\|_{op}$ as intended.

- **HS:** We'll alternatively write:

$$\|M\|_{HS} = \sqrt{\sum_{i=1}^n \sum_{j=1}^k M_{ij}^2} = \sqrt{\sum_{i=1}^n \|Me_i\|_2^2}$$

We clearly get for every matrix M that:

$$\|M\|_{HS}^2 = \sum_{i=1}^n \|Me_i\|_2^2 \geq \sum_{i=1}^n (\|M^{-1}\|_{op})^{-1}$$

Now:

$$\|AB\|_{HS}^2 = \sum_{i=1}^n \|A(Be_i)\|_2^2 \leq \sum_{i=1}^n \|A\|_{op}^2 \|Be_i\|_2^2 = \|A\|_{op}^2 \|B\|_{HS}^2$$

It remains to prove that $\|M\|_{op} \leq \|M\|_{HS}$ for each matrix M . The SVD decomposition of M yields orthonormal bases $u_1 \dots u_n$ and $v_1 \dots v_n$ s.t. $Mx = \sum_{i=1}^r \sigma_i v_i u_i^T \cdot x$ (where r is the rank). Now, assuming $r \neq 0$:

$$\|M\|_{HS}^2 = \sum_{i=1}^n \|Me_i\|_2^2 = \sum_{i=1}^n \left\| \sum_{i=1}^r \sigma_i v_i u_i^T \cdot e_i \right\|^2 = m \sum_{i=1}^n \sigma_i^2 \underbrace{\|v_i u_i^T\|_1}_1 \geq r \cdot \max_i \sigma_i^2 = m \|M\|_{op}^2 \geq \|M\|_{op}^2$$

Otherwise $r = 0$ and then $\|M\|_{HS} = \|M\|_{op} = 0$ and we finished. Finally $\|AB\|_{HS}^2 \leq \|A\|_{op}^2 \|B\|_{HS}^2 \leq \|A\|_{HS}^2 \|B\|_{HS}^2$ ergo $\|AB\|_{HS} \leq \|A\|_{HS} \|B\|_{HS}$ as intended. ■

2. Let $A \in \leftrightarrow (\mathbb{R}^n, \mathbb{R}^m)$ a linear operator with $u_1 \dots u_n$ orthonormal basis for $\mathbb{R}^n$. We'll prove $\|A\|_{HS}^2 = \sum_{i=1}^n \|Mu_i\|^2$.

Proof. For $u_i = e_i$ it follows from the definition. Otherwise, there exists some unitary matrix U s.t. $Ue_i = u_i$. Now $\|Mu_i\| = \|MUe_i\| = \|Me_i\|$ (as composing by a unitary matrix does not change a vector's norm). This finishes the proof. ■

3. We'll show that the parallelogram law holds for every norm induced from an inner product structure.

Proof. Consider $N(x) = \sqrt{\langle x | x \rangle}$. Then:

$$\begin{aligned} N(x+y)^2 + N(x-y)^2 &= \langle x+y | x+y \rangle + \langle x-y | x-y \rangle \\ &= \langle x | x+y \rangle + \langle y | x+y \rangle + \langle x | x-y \rangle - \langle y | x-y \rangle \\ &= \langle x | x \rangle + \langle x | y \rangle + \langle y | x \rangle + \langle y | y \rangle \\ &\quad + \langle x | x \rangle - \langle x | y \rangle - \langle y | x \rangle + \langle y | y \rangle \\ &= 2\langle x | x \rangle + 2\langle y | y \rangle + \cancel{2\langle x | y \rangle} - \cancel{2\langle y | x \rangle} \\ &= 2N(x)^2 + 2N(y)^2 \end{aligned}$$

As intended. ■

4. Consider $D \subset M_{2 \times 2}(\mathbb{R})$ be the set of diagonalizable matrices. Then:

- **Closure:** consider some $A \in M_{2 \times 2}(\mathbb{R})$. If A is diagonalizable, we finished. Otherwise, it's similar to Jordan $J_2(\lambda)$. Now the sequence of $J_\varepsilon(\lambda) := \begin{pmatrix} \lambda & 1 \\ \varepsilon & \lambda \end{pmatrix}$ for $\varepsilon \rightarrow 0$ clearly converge to $J_2(\lambda)$. Moreover, $J_\varepsilon(\lambda)$ are diagonalizable matrices as their characteristic polynomial is $(\lambda - x)^2 - 1 \cdot \varepsilon = \lambda^2 - 2\lambda x + x^2 + \varepsilon = 0$ so:

$$x = \frac{2\lambda \pm \sqrt{4 - 4\lambda^2\varepsilon}}{2}$$

We got two distinct eigenvalues for $J_\varepsilon(\lambda)$ hence it's diagonalizable. Now when $A = UJ_2(\lambda)U^{-1}$ we know that $A_\varepsilon := UJ_\varepsilon(\lambda)U^{-1}$ for $\varepsilon = \frac{1}{n}$ is a sequence of diagonalizable matrices. Moreover, it approaches A since:

$$\begin{aligned} \|A - A_\varepsilon\|_{\text{op}} &= \|UJ_2(\lambda)U^{-1} - UJ_\varepsilon(\lambda)U^{-1}\|_{\text{op}} = \|U(J_2(\lambda) - J_\varepsilon(\lambda))U^{-1}\|_{\text{op}} \\ &\leq \underbrace{\|U\|_{\text{op}} \|U^{-1}\|_{\text{op}}}_1 \|J_2(\lambda) - J_\varepsilon(\lambda)\|_{\text{op}} \\ &= \left\| \begin{pmatrix} 0 & 0 \\ \varepsilon & 0 \end{pmatrix} \right\|_{\text{op}} \leq \left\| \begin{pmatrix} 0 & 0 \\ \varepsilon & 0 \end{pmatrix} \right\|_{\text{HS}} = \varepsilon \rightarrow 0 \end{aligned}$$

We got that A is a limit point of a sequence in D . Therefore $A \in \text{Cl } D$. Finally $\text{Cl } D = M_{2 \times 2}(\mathbb{R})$.

- **Interior:** the complement of D is dense too. The proof is similar, except we're using $A_\varepsilon = U \begin{pmatrix} \lambda_1 & \varepsilon \\ 0 & \lambda_2 \end{pmatrix} U^{-1}$ instead (when $A \sim \text{diag}(\lambda_1, \lambda_2)$). Since $\text{Cl } X \setminus D = X$ then $\text{Int } D = X \setminus \text{Cl}(X \setminus D) = X \setminus X = \emptyset$.
- **Boundary:** clearly $\partial D = \text{Cl } D \setminus \text{Int } D = M_{2 \times 2}(\mathbb{R})$.

5. Directly from the definition.

..... (2)

1. Suppose for two norms N_1, N_2 the limit $\lim_{x \rightarrow 0} \frac{N_2(x)}{N_1(x)} = L$ exists. We'll prove $N_2 = L \cdot N_1$.

Proof. Let $x \in \mathbb{R}^d$. If $x = 0$, we finished. Otherwise, for each $\hat{\varepsilon}$ there is some δ s.t. $-\hat{\varepsilon} + L \leq \frac{N_2(x)}{N_1(x)} \leq L + \hat{\varepsilon}$. Let $\varepsilon > 0$ and consider $\hat{\varepsilon} = \frac{\varepsilon}{N_1(x)}$. We deduce from above that $|N_2(x) - L \cdot N_1(x)| \leq \varepsilon$ for all $\varepsilon \in \mathbb{R}$. Hence $N_2(x) - L \cdot N_1(x) = 0$ for each $x \in \mathbb{R}^d$, and we got $N_1 = L \cdot N_2$. ■

2. Let $X \subset \mathbb{R}^n$ be a path-connected set, and x_0 an accumulation point of X . Let $f: X \rightarrow \mathbb{R}$. We'll prove that $\lim_{x \rightarrow x_0} f(x) = y$ iff for any $\gamma: (0, 1) \rightarrow X \setminus \{x_0\}$ continues with $\lim_{t \rightarrow 0^+} \gamma(t) = x_0$, we get that $\lim_{t \rightarrow 0^+} f(\gamma(t)) = y$.

Proof. $\Rightarrow$ Suppose that $\lim_{x \rightarrow x_0} f(x) = y$. Let γ be such path. Let $0 < t_n \rightarrow 0$ be a sequence in $\mathbb{R}$ (WOLOG bounded by 1). Now, $x_n := \gamma(t_n)$ is a sequence in $X \setminus \{x_0\}$, and because $\lim_{t \rightarrow 0^+} \gamma(t) = x_0$ we know that $x_0 \neq x_n \rightarrow x_0$. Since $\lim_{x \rightarrow x_0} f(x) = y$ and $x_0 \neq x_n \rightarrow x_0$, we know that $f(x_n) \rightarrow y$ too. Now for each t_n approaching 0 from above we got that $f(\gamma(t_n)) = f(x_n) \rightarrow y$ and from Heine we get that $\lim_{t \rightarrow 0^+} f(\gamma(t)) = y$.

$\Leftarrow$ Suppose that any such γ has the property $\lim_{t \rightarrow 0^+} f(\gamma(t)) = y$. Consider some $x_n \rightarrow x_0$ with $x_n \neq x_0$. As each $x_i, x_{i+1} \in X$ and X is path-connected there exists a continues path $\gamma_i: [0, 1] \rightarrow X$ s.t. $\gamma_i(0) = x_i \wedge \gamma_i(1) = x_{i+1}$. Now consider the infinite sum $\gamma := \sum_{i=1}^{\infty} \gamma_i$ defined as:

$$\gamma := \sum_{i=1}^{\infty} \gamma_i := \lambda x \in (0, 1). \begin{cases} \gamma_1(2x - 1) & x \in [\frac{1}{2}, 1] \\ \gamma_2(4x - 1) & x \in [\frac{1}{4}, \frac{1}{2}] \\ \vdots & \vdots \\ \gamma_n(2^n x - 1) & x \in [\frac{1}{2^n}, \frac{1}{2^{n-1}}] \end{cases}$$

Note that this is well-defined as if $x = \frac{1}{2^n}$ then $\gamma_n(2^n x - 1) = \gamma_n(0) = \gamma_{n+1}(1) = \gamma_{n+1}(2^{n+1}x - 1)$. For this reason γ is also continues and for each $n \neq 0, 1$ we get that $\gamma(2^{-n}) = x_n$. Since γ is continues and we have partial limit of x_0 , γ must converge to x_0 for $t \rightarrow 0^+$. Now $\lim_{t \rightarrow 0^+} f(\gamma(t)) = y$ therefore $f(\gamma(\frac{1}{2^n})) = f(x_n)$ must converge to y too, as intended. ■

3. Let $E \subset X$. We'll prove that the indicator $\mathbb{1}_E$ is discontinuous on $x \in X$ iff $x \in \partial E$.

Proof. If $\partial E = \emptyset$, we finished. Otherwise:

$\implies$ Suppose that $\mathbb{1}_E$ is discontinuous at x . Then either the limit does not exist or

$\Leftarrow$ Suppose that $x \in \partial E = \text{Cl } E \cap \text{Cl } (X \setminus E)$. Then $x \in \text{Cl } E$ therefore there exists a sequence $x_n \rightarrow x$ s.t. $x_n \in E$. In addition there is a sequence $y_n \rightarrow x$ s.t. $y_n \in X \setminus E$. By definition $f(y_n) = 0 \wedge f(x_n) = 1$ therefore from Heine the limit $\lim_{y \rightarrow x} f(y)$ does not exist. ■

4. (a) Consider the function $f(x, y) = \frac{x+y}{\log(x^2+y^2)}$. We'll prove when $(x, y) \rightarrow (0, 0)$ then $f(x, y) \rightarrow 0$. From AMGM we know that $x^2 + y^2 \geq 2xy$. Since $\log$ monotonically increases, we get that $\log(x^2 + y^2) \geq \log(2xy)$, and $|x^2| \leq |\log x|$ for small enough x ($x < 0.5$) and therefore:

$$|f(x, y)| = \left| \frac{x+y}{\log(x^2+y^2)} \right| \leq \left| \frac{x+y}{\log(2xy)} \right| = \left| \frac{x+y}{\log 2x + \log 2y} \right| \stackrel{!}{\leq} \left| \frac{x+y}{4x^2+4y^2} \right| \rightarrow 0$$

($\stackrel{!}{\leq}$ is true only once (x, y) reaches certain bound)

6. Let $\emptyset \neq A \subset \mathbb{R}^d$. We'll prove that if any continuous $f: A \rightarrow \mathbb{R}$ is bounded, then $f: A \rightarrow \mathbb{R}$ attains maximum & minimum.

Proof. $\implies$ AFSOC A is not closed and bounded.

- If A is not bounded, then for $f(x) = x$ the identity, f is not bounded although it is continuous which is a contradiction.
- If A is not closed, then $\text{Cl } A \setminus A \neq \emptyset$. Therefore there is some point $x_0 \in \text{Cl } A \setminus A$, which implies the existence of a sequence $x_n \rightarrow x_0$ s.t. $x_n \in A$. WOLOG $\|x_n\|$ monotonically increases (otherwise we can always choose a monotonic subsequence of x_n instead, and for monotonically decreasing sequences the solution is similar). Now define f as follows:

$$f(x) = \begin{cases} \frac{1}{n-1} + (\|x\| - \frac{1}{n}) & \|x\| \in [\|x_{n-1}\|, \|x_n\|] \\ \inf A + \|x\| & \|x\| \in [\inf A, x_1] \\ \|x_0 + (x - v)\| & \|x\| \in [\|x_0\|, \sup \|A\|] \end{cases}$$

(Where v is some $v \in \mathbb{R}^d$ s.t. $\|v\| = \sup A$) WOLOG A is bounded otherwise we proved a contradiction. From the definition of linear interpolation, and because $\|x_0\| = \sup \|x_n\|$, it's clear that f is continuous. Moreover, $f(x_n) = \frac{1}{n}$ and therefore f is not bounded.

Finally A is closed and bounded and from Heine-Borel it's also compact. Therefore $f(A)$ is compact too and we finished.

$\Leftarrow$ If each $f: A \rightarrow \mathbb{R}$ attains a maximum and minimum, it's in particular bounded. ■

7. Suppose $\emptyset \neq A, B \subset \mathbb{R}^d$ are closed and disjoint.

(a) We'll prove $x \mapsto d(x, A)$ is continuous and $d(x, A) = 0$ iff $x \in A$.

Proof. If $x \in A$ clearly $d(x, A) = 0$. Otherwise since A is closed there is a basic open neighborhood of x s.t. $x \in U = B(x, \varepsilon)$ and $U \cap A = \emptyset$. Now clearly $d(x, A) \geq \varepsilon > 0$ and we finished.

We'll proceed to prove that $x \mapsto d(x, A)$ is continuous. For each $\varepsilon > 0$ exists N s.t. $d(x_{n>N}, x) < \varepsilon$. Let $\varepsilon > 0$ and choose such N . Now:

$$\begin{aligned} d(x, A) &\leq d(x, x_n) + d(x_n, A) &\implies & d(x, A) - d(x_n, A) \leq \varepsilon \\ d(x_{n_k}, A) &\leq d(x_n, x) + d(x, A) &\implies & d(x_n, A) - d(x, A) \leq \varepsilon \end{aligned}$$

Therefore $|d(x_n, A) - d(x, A)| \leq \varepsilon$ for $n > N$. By definition $d(x_n, A) \rightarrow d(x, A)$ as intended. ■

(b) Clearly exists such Urysohn function because $\mathbb{R}^d$ is completely normal. See topology exercises.

. (3)

1. Let $\gamma, \eta: \mathbb{R} \rightarrow \mathbb{R}^n$ to differentiable functions. We'll prove that:

$$\frac{d}{dt} \langle \gamma(t) | \eta(t) \rangle = \langle \gamma'(t) | \eta(t) \rangle + \langle \gamma(t) | \eta'(t) \rangle$$

Proof. Denote $h(t) = \gamma(t)^T \eta(t)$. Then:

$$\begin{aligned} h(t+s) - h(t) &= \langle \gamma(t+s) | \eta(t+s) \rangle - \langle \gamma(t) | \eta(t) \rangle \\ &= \langle \gamma(t+s) | \eta(t+s) \rangle - \langle \gamma(t+s) | \eta(t) \rangle - \langle \gamma(t) | \eta(t) \rangle + \langle \gamma(t+s) | \eta(t) \rangle \\ &= \langle \gamma(t+s) | \eta(t+s) - \eta(t) \rangle + \langle \gamma(t+s) - \gamma(t) | \eta(t) \rangle \end{aligned}$$

And by definition:

$$\begin{aligned} \frac{d}{dt} h(t) &= \lim_{s \rightarrow 0} \frac{h(t+s) - h(t)}{s} = \lim_{s \rightarrow 0} \frac{\langle \gamma(t+s) | \eta(t+s) - \eta(t) \rangle + \langle \gamma(t+s) - \gamma(t) | \eta(t) \rangle}{s} \\ &= \left(\lim_{s \rightarrow 0} \gamma(t+s) \right)^T \left(\lim_{s \rightarrow 0} \frac{\eta(t+s) - \eta(t)}{s} \right) + \left(\lim_{s \rightarrow 0} \frac{\gamma(t+s) - \gamma(t)}{s} \right)^T \eta(t) \\ &= \gamma(t)^T \eta'(t) + \gamma'(t)^T \eta(t) = \langle \gamma(t) | \eta'(t) \rangle + \langle \gamma'(t) | \eta(t) \rangle \end{aligned}$$

And if $\|\gamma(t)\|$ is constant $\sqrt{c}$, then $c = \|\gamma(t)\|^2 = \langle \gamma(t) | \gamma(t) \rangle$ is constant too, which happens if and only if its derivative is 0, and by the theorem above:

$$0 = \langle \gamma | \gamma' \rangle + \langle \gamma' | \gamma \rangle = 2 \langle \gamma' | \gamma \rangle \iff \langle \gamma' | \gamma \rangle = 0 \iff \gamma' \perp \gamma$$

■

2. Let $f, g: \mathbb{R}^n \rightarrow \mathbb{R}$ by differentiable functions. We'll prove $f \cdot g$ is differentiable too, and find $\nabla(f \cdot g)$.

Proof. Denote $h = f \cdot g$. Then:

$$\begin{aligned} h(x+v) - h(x) &= f(x+v)g(x+v) - f(x)g(x) \\ &= f(x+v)g(x+v) - f(x+v)g(x) - f(x)g(x) + f(x+v)g(x) \\ &= f(x+v)(g(x+v) - g(x)) + g(x)(f(x+v) - f(x)) \end{aligned}$$

Consider the linear transformation $A_x := f(x)\nabla(g) + g(x)\nabla(f)$. Now (no need for $|\cdot|$ since $|P| \rightarrow 0$ iff $p \rightarrow 0$):

$$\begin{aligned} \lim_{v \rightarrow 0} \frac{h(x+v) - h(x) - Av}{\|v\|} &= \lim_{v \rightarrow 0} \frac{f(x+v)(g(x+v) - g(x)) + g(x)(f(x+v) - f(x)) - Av}{\|v\|} \\ &= \lim_{v \rightarrow 0} f(x+v) \frac{g(x+v) - g(x)}{\|v\|} \end{aligned}$$

■

3.

4. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a differentiable function. Suppose $W \subset \mathbb{R}^m$ is a linear space s.t. $\text{Im } f \subset W$. We'll prove $\text{Im}(D_f x) \subset W$.

Proof. Let $x \in \mathbb{R}^n$. Ad absurdum there exists $y \in \mathbb{R}^n$ s.t. $(D_f x)y \notin W$. Then for all $\varepsilon \in \mathbb{R}$ we know that $(D_f x)(\varepsilon \cdot y) \notin W$ from the linearity of $D_f x$. Now notice that:

$$0 = \lim_{x' \rightarrow x} \frac{\|f(x') - f(x) - (D_f x)(x - x_0)\|}{\|x - x'\|} = \lim_{y' \rightarrow 0} \frac{\|f(x+y') - f(x) - (D_f x)(y')\|}{\|y'\|}$$

And in particular, for the subsequence $\varepsilon_n = \frac{1}{n}$ and $y_n = \varepsilon_n \cdot y$. Then:

$$0 = \lim_{n \rightarrow \infty} \frac{\| \overbrace{f(x+y_n) - f(x)}^{:= p_n \in W} - \overbrace{(D_f x)(y_n)}^{:= q_n \notin W} \|}{\|y_n\|} = \|y\| \cdot \lim_{n \rightarrow \infty} \frac{\|p_n - q_n\|}{\frac{1}{n}} = \lim_{n \rightarrow \infty} n \|p_n - q_n\| = \lim_{n \rightarrow \infty} \|np_n - (D_f x)(y)\|$$

Therefore $np_n \xrightarrow{n \rightarrow \infty} (D_f x)(y) \notin W$, although $np_n \in W$ (linearity) so $\lim_{n \rightarrow \infty} np_n \in W$ (W is a closed subset of $\mathbb{R}^m$, as a linear space), which forms a contradiction.

■

5. Suppose that $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable at the origin and $f(\lambda x) = \lambda f(x)$ for all $\lambda \in \mathbb{R}$. We'll prove that f is linear.

Proof. Clearly $f(0) = 0$ since $f(0) = f(0 \cdot 0) = 0f(0) = 0$. Denote $D_f := D_f 0$. Now for all x, y we know that $\left\| \frac{f(0) - f(x+y) - D_f \cdot (x+y)}{x+y} \right\| \rightarrow 0$ as $x+y$ approaches 0, and considering $f(0) = 0$ and the linearity of D_f we get that $\left\| \frac{f(x+y) - ((D_f)x + (D_f)y)}{x+y} \right\| \rightarrow 0$. In particular for all of the λ -multiplicities of x, y . It immediately follows from linearity of D_f and from the fact that $f(\lambda x) = \lambda f(x)$ that:

$$\begin{aligned} 0 &= \lim_{x+y \rightarrow 0} \left\| \frac{f(x+y) - ((D_f)x + (D_f)y)}{x+y} \right\| \stackrel{!}{=} \lim_{\lambda \rightarrow 0} \left\| \frac{\lambda f((x+y)) - \lambda ((D_f)x + (D_f)y)}{\lambda(x+y)} \right\| \\ &= \left\| \frac{f((x+y)) - ((D_f)x + (D_f)y)}{(x+y)} \right\| \end{aligned}$$

Where $\stackrel{!}{=}$ is true as we're taking a subsequence of $x+y \rightarrow 0$. It implies that $f(x+y) = (D_f)x + (D_f)y$ therefore f is linear. ■

6. Consider $F(A) = A^2$ ($F: M_{n \times n}(\mathbb{R}) \leftrightarrow$). We'll show by definition that F is differentiable with $[D_f A](H) = AH + HA$.

Proof. We need to prove that $\lim_{H \rightarrow 0} \frac{\|f(A+H) - f(A) - [D_f A](H)\|}{\|H\|} = 0$. By declaring $[D_f A](H) = AH + HA$ this is clear since:

$$\frac{\|(A+H)^2 - A^2 - AH - HA\|}{\|A - H\|} = \frac{\|H^2\|}{\|A - H\|}$$

For all $A \neq 0$, A has a singular value greater than 0 and then $\lim_{H \rightarrow 0} \|A - H\| = \|A\| \neq 0$. In this case:

$$\lim_{H \rightarrow 0} \frac{\|f(A+H) - f(A) - [D_f A](A)\|}{\|H\|} = \lim_{H \rightarrow 0} \frac{\|H\|}{\|A - H\|} = \frac{0}{\|A\|} = 0$$

If $A = 0$ then we get the limit: [all norms are equal, so we'll use the operator norm]

$$\lim_{H \rightarrow 0} \frac{\|H^2\|}{\|A - H\|} = \lim_{H \rightarrow 0} \frac{\|H\|^2}{\|H\|} = \lim_{H \rightarrow 0} \|H\| = 0$$

Either way the limit approaches 0 as intended. ■